

## **Tutorial 7 – Template**

1. In a banking system, you are asked to develop a template function in C++ which is a generic account balance calculator. The template is called **calculateInterest**, that takes balance and interest rate, then it returns the updated balance after interest is added basing on the rate.

The main function will prompt user to enter the account balance and interest rate the bank offers at that time. The account balance will be displayed before and after the calculation of interest.

2. Use C++ to develop a template function **findAverage** which has a single parameter. It is used to find the average mark of 5 subjects of a student. In the main function, declare an array variable **marks** which is initialized with the 5 marks obtained by a student for 5 subjects. It will use the above function to find and display the average of these 5 marks.

3. Create a main function in C++, prompt the user to enter the two names of book prize winner in A category and B category. It will use a template function to display an announcement message as below:-

“The book prize winners for A and B categories are Gary Heinz and Flora Pat.”

4. Write a template function in C++ to check the price of 2 items which are of structure type **Item** which contains item code and price. Declare the structure **Item** and initialize 2 items in the main function and print the item code with cheaper price.

5. Develop a template function in C++ to return the minimum value of 3 given values of different types. In the main function, perform the template function to find and display the minimum value of 3 integer values and 3 float type values.

6. Write a template function **increaseSalary** which increases all the employee salaries by a given percentage.

In the main function, a structure **Employee** with id, name and salary will be defined. Then declare an array variable **emp** of the Employee structure and initialized with 3 sets of employee information. The employee information should be displayed before and after the increment of salary by certain percentage.